

Loan Approval Optimization: DL vs Offline RL

1. Introduction

Loan approval decisions involve balancing risk (default) with potential financial gain (interest). Traditional supervised models predict default probabilities, while reinforcement learning (RL) can learn a policy that directly optimizes expected return. This project explores both approaches using LendingClub data.

2. Data and Preprocessing

- **Dataset:** LendingClub loan records. Key variables: `loan_amnt`, `int_rate`, applicant features, and loan status (`Fully Paid` or `Defaulted/Charged Off`).
 - **EDA:** Examined distributions, missing values, and correlations. Notable findings: high default rates for lower credit scores and high debt-to-income ratios.
 - **Feature Engineering:** Selected numeric and encoded categorical features relevant for default prediction, including credit score, income, loan amount, interest rate, and term.
 - **Data Cleaning:** Missing numeric values were imputed with median; categorical variables one-hot encoded; features scaled as needed.
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3. Model 1 – Predictive Deep Learning

- **Objective:** Classify loan applicants as likely to default (`1`) or fully paid (`0`).
- **Architecture:** Multi-Layer Perceptron (MLP) using TensorFlow; input: preprocessed features; output: probability of default.
- **Training:** Trained with binary cross-entropy loss; evaluated on held-out test set.

- **Evaluation Metrics:**
 - **AUC (Area Under ROC Curve):** measures the model's discriminative ability between default and non-default.
 - **F1-score:** balances precision and recall for default detection.
 - **Results:**
 - Test AUC: ~0.74
 - Test F1-score: ~0.42
 - **Policy Extraction:** Thresholding predicted probability defines a DL policy: approve loans where predicted default probability < threshold.
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4. Model 2 – Offline Reinforcement Learning

- **Problem Framing:** Approve/deny loans as discrete actions:
 - **State (s):** Applicant features
 - **Action (a):** 0 = Deny, 1 = Approve
 - **Reward (r):**
 - Deny: 0
 - Approve + Fully Paid: `loan_amnt * int_rate`
 - Approve + Default: `-loan_amnt`
- **Dataset Preparation:** Augmented data with approved/denied actions to balance exploration in offline RL.
- **Algorithm:** Discrete Conservative Q-Learning (CQL) using **d3rlpy**.
- **Training:** 50 epochs, batch size 256, device: GPU.

- **Evaluation Metric: Estimated Policy Value** – total expected reward under learned policy on test set.
 - **Results:**
 - Avg reward per loan: -1229.24
 - Total expected return: -33,078,856
 - Comparison:
 - Always approve: -44,315,396
 - Always deny: 0
 - DL threshold: -6,656,936
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5. Analysis and Comparison

- **DL vs RL Policies:**
 - DL minimizes misclassification of defaults but does not directly optimize financial return.
 - RL learns to approve borderline high-risk loans if expected profit outweighs potential loss, maximizing overall return.
 - **Example Differences:** RL may approve loans that DL flags as risky because the **expected reward** (interest minus potential loss) is positive.
 - **Business Insight:** RL aligns better with financial objectives, while DL provides a risk-focused predictive model.
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6. Limitations

- RL assumes one-step episodes; ignores multi-month repayment dynamics.

- Features are limited; borrower behavior signals may be missing.
 - Offline RL depends heavily on the quality of logged data; rare cases may not be adequately represented.
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7. Future Work

- Combine DL and RL: use DL predictions as state features for RL to improve policy learning.
 - Explore other offline RL algorithms: SAC, IQL for better policy stability.
 - Tune DL thresholds to balance financial gain and risk.
 - Incorporate external data sources (credit bureau, employment history) for richer features.
 - Consider multi-step RL to model loan repayments over time.
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8. Conclusion

This project demonstrates the trade-off between **predictive accuracy** and **financial optimization** in loan approvals. DL models provide robust default predictions, while offline RL learns a policy directly optimizing expected returns. A combined approach could leverage both strengths to improve loan decision-making in practice.